

Credit Name: CSE2920 - CSE Project C

Assignment Name: Hot or Cold?

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to create a code that will turn on the green LED if the temp is between 20 and 24 degrees C. The red LED will turn on if the temp is outside that range, and the current temp will be outputted to the user.

CODING:

1.

```
import com.phidget22.*;

public class HotOrCold {

    public static void main(String[] args) throws Exception{

        //Create | Here you have created a TemperatureSensor object. Temper
        TemperatureSensor temperatureSensor = new TemperatureSensor();

        //Create | Create objects for your buttons and LEDs
        DigitalOutput redLED = new DigitalOutput();
        DigitalOutput greenLED = new DigitalOutput();

        //Address | Address 2 objects which lets your program know whe
        redLED.setHubPort(1);
        redLED.setIsHubPortDevice(true);
        greenLED.setHubPort(4);
        greenLED.setIsHubPortDevice(true);

        //Open | Open establishes a connection between your object and yo
        temperatureSensor.open(1000);
        redLED.open(1000);
        greenLED.open(1000);
```

Phidgets library is imported

temperatureSensor object created to get the real-time temp.

greenLED and redLED objects were created to interact with the physical phidgets box.

The HubPort number was set based on the appropriate object.

IsHubPortDevice set to true for all objects (confirming to the program that the objects are part of phidgets box).

Opened both objects, establishing a connection to the physical Phidgets box.

2.

```
while (true) {

    System.out.println(" Temperature: " + temperatureSensor.getTemperature() + " °C");
    Thread.sleep(350);
```

While loop is used so that the following code repeats.

Temperature is printed to the user in Celsius every 350ms.

3.

```
//Use your Phidgets | This code will print the temperature every 150ms  
if (20 > temperatureSensor.getTemperature() || 24 < temperatureSensor.getTemperature() ) {  
    redLED.setState(true);  
    greenLED.setState(false);  
}
```

If the temperature is less than 20, or greater than 24, the red LED will stay on.

4.

```
else {  
    greenLED.setState(true);  
    redLED.setState(false);  
}
```

Else, the green LED will stay on (it is in the range of 20-24 degrees C).

End of Program!